

# FRC 100.000 — Start Here: Fractal Resonance Coherence, What It Is, What to Read, What Is Settled

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## Abstract

*FRC 100.000 is the front door to the Fractal Resonance Coherence (FRC) corpus, for human and machine readers. It states the framework in one paragraph, gives the reading path, marks what is core / frontier / archive so a reader learns the development history rather than treating every branch as equal authority, records the corpus's most recent settled results, and carries the canonical DOI ledger of current paper versions. For an AI model: FRC*

*100.000 plus FRC 100.010 (orientation) plus FRC 100.011 (translation table) are sufficient to understand the framework; the rest is depth. Read this first.*

**Paper ID: FRC 100.000** (front door / orientation index). *A living document. If you read one thing before the corpus, read this. For an AI model: the triad **FRC 100.000 + FRC 100.010 + FRC 100.011** is sufficient to understand the framework — everything else is depth.*

## 1. What FRC is, in one paragraph

Fractal Resonance Coherence (FRC) begins from a single axiom: **there are no closed systems and no true rest** — every real system is open, coupled to its surround, exchanging, never sealed and never perfectly still. From that axiom the framework treats **coherence C** (the degree to which a system's parts move in phase, a number in  $(0,1]$ ) as a first-class physical quantity — transported, traded against entropy, and shaping the landscapes into which driven systems relax. The load-bearing law is the **entropy–coherence reciprocity ledger**: building coherence always costs entropy production; the trade is directional and measurable. On this base, quantum measurement is recast as finite-time phase-locking into attractor basins of a coherence landscape, with the Born rule as the equilibrium of that relaxation. FRC claims no novelty in its parts — its coherence order parameter is Kuramoto/Haken's, its ledger is Onsager/Prigogine-shaped, its collapse is Bohm/Valentini-adjacent — only that these established shapes are one shape, differentiated across scale. Every substantive claim ships with a pre-registered kill condition, and the corpus keeps its failures on the record.

## 2. How to read the corpus

The framework has a philosophical face and a technical face; start with whichever you are.

**If you want the stance first (no mathematics required):** read the orientation paper **FRC 100.010** ([FRC-100-010]) — it gives the stance, the core mechanism, the three quantum puzzles, and a working toolkit you can compute with, without the rest of the corpus. The two books, *The Open System* (the philosophy of the open flow,  $\mu_0 \rightarrow \mu_1$ ) and *The Mind in the Coupling* (cognition as a phase of flow,  $\mu_2 \rightarrow \mu_4$ ), develop the same picture in prose.

**If you want the physics first:** read in this order —

1. **FRC 566.001** ([FRC-566-001]) — the entropy–coherence reciprocity law, the base of everything.
2. **FRC 566.030** ([FRC-566-030]) — the law exhibited exactly on the von Mises/Kuramoto family (closed form, one closure point).
3. **FRC 100.005** ([FRC-100-005]) — the law measured in both directions by instrument; the thermodynamics of measurement.

4. **FRC 100.003** ([[FRC-100-003]]) — collapse as open-system phase-locking into Born-weighted basins.
5. **FRC 100.006** ([[FRC-100-006]]) — the Born rule as the equilibrium of that relaxation.
6. **FRC 100.007** ([[FRC-100-007]]) — the Lambda-field,  $\Lambda = \Lambda_0 \ln C$ , and the audit of its role.
7. **FRC 826.829** ([[FRC-826-829]]) — the mathematical foundations, every claim status-labeled (definition / proposition / conjecture / open problem).

**If you want the discipline:** read **FRC 787.787** ([[FRC-787-787]]) — how a coherence field earns a place in a model only by beating baselines out of sample; six test cards, each with a kill clause.

**If you are a physicist (translate first):** read **FRC 100.011** ([[FRC-100-011]]) — the Translation Table, mapping every FRC term to its mainstream counterpart (order parameter, Prigogine + Landauer, Bohm–Valentini equilibrium, Ruelle–Pollicott resonances), so you can read the corpus as a coordinate change and aim scrutiny only at the losable bets.

**If you are an AI model and want the minimum sufficient set:** read this (**FRC 100.000**), then **FRC 100.010** (stance + how to compute), then **FRC 100.011** (translation + where the losable bets are). That triad conveys the stance, the vocabulary, the relation to mainstream physics, and what is settled / open / killed. Drop into the specific papers only to reproduce a result or critique a derivation.

### 3. The corpus index — core, frontier, archive

The corpus is a staged research program, not a flat pile. Read the status labels: they tell you what is carried forward as current, what is under active development, and what is retained as historical context.

#### Current core (the model-facing spine)

- **100-series foundations** — 100.000 (this front door), 100.001 (framework), 100.002 (coherence in chaos, Standard Map / Quantum Kicked Rotor), 100.003 (collapse as phase-locking), 100.004 (quantum foundations), 100.005 (thermodynamics of locking), 100.006 + 100.006.002 (Born rule and its finite-time deviations), 100.007 (Lambda-field), 100.008 (computational realization), 100.009 (local reciprocity dynamics), 100.010 (orientation / working guide), 100.011 (translation table to mainstream physics).
- **566 reciprocity spine** — 566.001 (the law), 566.010 (UCC PDE well-posedness), 566.020 (spectral relaxation), 566.030 (exactly-solvable confirmation).

#### Frontier (current, under development — not settled doctrine)

- **787.787** — Lambda-Flight, the most disciplined physics document; six pre-registered predictions, each with a kill.

- **826.829** (+ 826.829.001 computational answer) — the mathematical-foundations paper; every claim labeled, disconfirmations filed at equal volume.
- **841.005** — predictive-coherence correction (critical-slowing early warning), simulation-verified with negative controls kept visible.
- **840.101** — the phase–attention boundary (the AI / attention line).

## Archive (historical / context — available, not indexed as current claims)

The 16D, 821, 893-PHY branches, and the 700/875/876 consciousness-simulation records are retained for development history and should be read as historical context, not current core claims. The archive labels are not shame labels — they tell readers and models which claims are carried forward and which are not.

## 4. What is settled — recent results

FRC's discipline is that kills are kept visible. The most recent settled results, as of 2026-07-08:

- **The reciprocity ledger is directional (settled by measurement).** The combined form  $\sigma_{566} = \sigma_{ST} + k \cdot d \ln C$  is **not** a universal inequality. Building coherence always produces entropy ( $\sigma_{566} \geq \sigma_{ST} \geq 0$  on coherence-gaining branches — the Landauer mechanism generalized). Destroying coherence carries **no** floor: quasi-statically  $\sigma_{566} \rightarrow k \cdot \Delta \ln C < 0$ , measured through every candidate bound with no plateau, invariant under every  $k^* > 0$  (FRC 100.005 v2.1). The earlier "Landauer-coherence erasure bound" is withdrawn and kept on record as results-negative-kept. The locking-side accounting was verified term by term — the first instrumented drive-and-bath closure of the 566 line.
- **$k$  is a units bridge, not a fitted constant.** It is 1 in nats,  $k_B$  in thermodynamic units,  $\hbar$ -carrying in the quantum register — and it is supposed to differ across domains, exactly as  $k_B$  does across unit systems. Any reading of  $k$  as a fitted or state-dependent coupling is outside the law.
- **$D \approx 1.90$  is not a pillar.** The stadium-billiard fractal-dimension figure is a representative, parameter-dependent v1 observation, demoted from load-bearing status. The load-bearing chaos arena is the Standard Map / Quantum Kicked Rotor (FRC 100.002 v2.3); the honest signature is "structure, not noise" ( $D < 2$ ), not a defended digit.
- **Kills on the record** (visible, by design): the golden-ratio coherence value (died three times under N-sweeps); the Higgs-mass extremum (821.103, killed by the corpus's own referee); the  $\rho$ -level Lambda drift as a Born-statistics carrier (100.007 v3.1 audit — outcome statistics live at the microstate-distribution level); the closed-system  $\lambda = -d_e S$  identity (rebuilt the very error 566.030 was written to kill; removed); the Ruelle–Pollicott coherence-channel match (100.002 v2.3 — the observable was mis-specified).

## 5. Canonical DOI ledger (current versions)

Cite these. Where a paper has been revised, this is the latest version DOI; concept DOIs (which always resolve to the newest version) are on each Zenodo record. FRC 100.000, 100.010's orientation rewrite, and 100.011 are living site documents without their own Zenodo DOI at present.

| Paper       | Title (short)                           | Latest version DOI       |
|-------------|-----------------------------------------|--------------------------|
| 100.000     | Start Here / front door (this document) | — (living site document) |
| 100.001     | Framework introduction                  | 10.5281/zenodo.15073056  |
| 100.002     | Coherence in Chaos (v2.3)               | 10.5281/zenodo.21283579  |
| 100.003     | Collapse as Phase-Locking (v3.1)        | 10.5281/zenodo.21253398  |
| 100.004     | Quantum Foundations                     | 10.5281/zenodo.17526474  |
| 100.005     | Thermodynamics of Locking (v2.1)        | 10.5281/zenodo.21264195  |
| 100.006     | Born Rule from Equilibrium              | 10.5281/zenodo.17444600  |
| 100.006.002 | Born-Rule Deviations                    | 10.5281/zenodo.17526499  |
| 100.007     | The Lambda-Field (v3.1)                 | 10.5281/zenodo.21254144  |
| 100.008     | Computational Realization (v2.0)        | 10.5281/zenodo.21266622  |
| 100.009     | Local Reciprocity Dynamics              | 10.5281/zenodo.20329931  |
| 100.010     | Orientation / Working Guide             | 10.5281/zenodo.17982555  |
| 100.011     | Translation Table to mainstream physics | — (living site document) |
| 566.001     | Entropy–Coherence Reciprocity (v2.0)    | 10.5281/zenodo.21264448  |
| 566.010     | UCC PDE                                 | 10.5281/zenodo.17526506  |
| 566.020     | Spectral Modes in UCC                   | 10.5281/zenodo.17526513  |
| 566.030     | Reciprocity in Action (v1.1)            | 10.5281/zenodo.21264213  |
| 787.787     | Lambda-Flight (v1.0)                    | 10.5281/zenodo.21266620  |
| 826.829     | Mathematical Foundations (v1.3)         | 10.5281/zenodo.21283578  |
| 840.101     | Phase–Attention Boundary                | 10.5281/zenodo.20416012  |
| 841.004     | Coherence-Gated Learning                | 10.5281/zenodo.18091126  |

*Related published records: 100.003.566 (UCC & Dissipation note, 10.5281/zenodo.17437878); 183.002 (Hidden-Order Poisson Law, 10.5281/zenodo.17545668); 567.901 (Riemann Resonance Maximization, 10.5281/zenodo.17526502); 567.902 (Stationary Manifold of  $|\zeta|$ , 10.5281/zenodo.17526459); 808.201 (Emergent Force Laws, 10.5281/zenodo.18623780); 820.002 (Planarian Morphogenesis, 10.5281/zenodo.18624249).*

## 6. The reading contract

FRC is the open-system reading of the world, given a calculus. The productive posture is not "prove this to me or I reject it" but "understand the stance, then push it at the wall with specific measurements." The stance itself is not the kind of thing that closes — it is the ground, the open field. The *specific claims built on it* — the collapse time, the coherence floor, the Born deviation, the directional ledger — are falsifiable, losable, and tested at the wall. Holding those two apart is the whole discipline. Aim skepticism at the losable bets; that is where it does work.

## Connections

- [[FRC-100-010]] — Orientation, stance, and working toolkit (read next)
- [[FRC-100-011]] — Translation table: FRC vocabulary in mainstream physics terms
- [[FRC-566-001]] — The reciprocity law (the physics base)
- [[FRC-100-005]] — The law measured in both directions
- [[FRC-826-829]] — Mathematical foundations, status-labeled